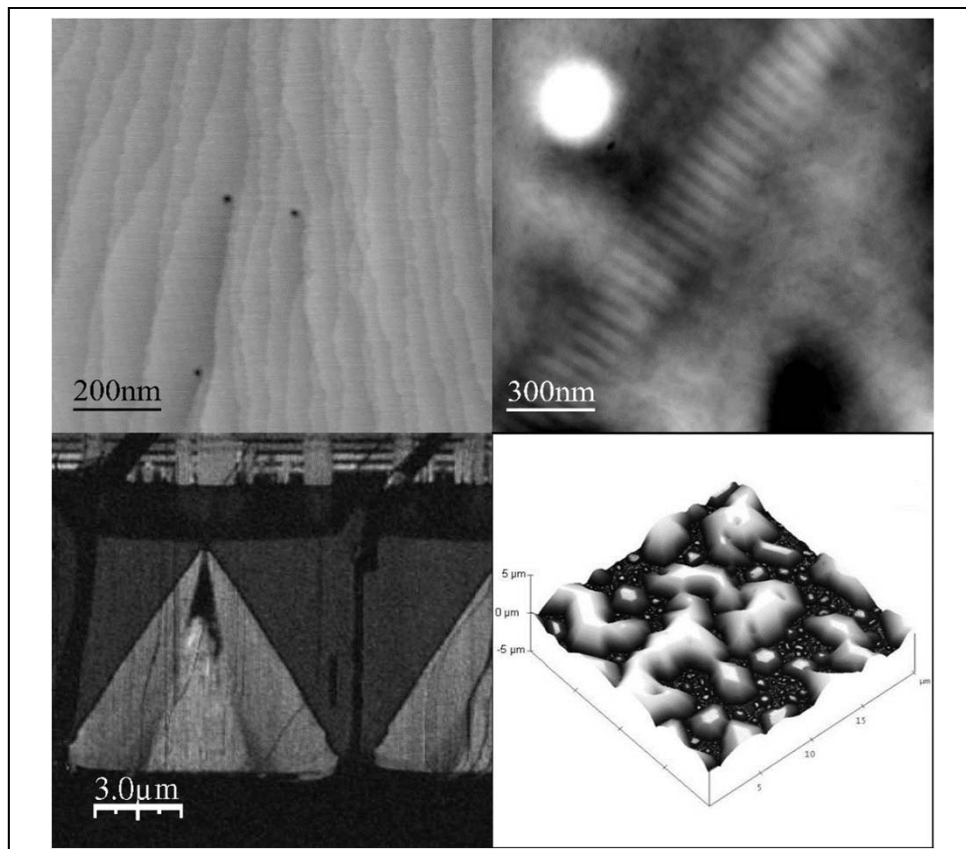


Micro- and Nanotechnology Enterprise
NE.01: Characterisation Techniques
Part 2: Scanning Probe Microscopy



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1. Introduction

This course introduces students to the practical aspects of scanning probe microscopy (SPM), with a significant focus on atomic force microscopy (AFM). Scanning tunneling microscopy (STM) and nearfield scanning optical microscope (SNOM) will also be briefly introduced.

By the end of the course, students should have sufficient knowledge to be able to (1) gauge whether different modes of SPM can be of use for their research project, (2) critically assess (their own or other people's) SPM data, and (3) understand the principles involved in optimization of SPM data acquisition, as a starting point for their own potential studies using this technique.

1.1 Useful resources

- DoITPoMS Teaching and learning package: “Atomic Force microscopy” <https://www.doitpoms.ac.uk/tlplib/afm/index.php> - Good introduction to the topic, developed in this Department in 2009.
- P. Eaton and P. West, “Atomic Force Microscopy”, Oxford : Oxford University Press, 2010 (Available in the MSM library) - Relatively recent, covers many of the topics in this course well.
- E. Meyer, H.J. Hug and R. Bennewitz, “Scanning Probe Microscopy: The Lab on a Tip”, London: Springer, 2004 (Available in the MSM library) - Intended as an introductory book for graduate students.
- P.C. Braga and R. Davide, “Atomic Force Microscopy: Biomedical Methods and Applications”, Totowa, NJ: Humana Press, 2004 (Available in the MSM library) - Basic principles of AFM and its uses in biomedical and life sciences.

1.2 A bit of historical context

Historically, the first scanning probe microscope was the STM. It was invented in 1981 by IBM researchers Gerd Binnig and Heinrich Rohrer, who were able to resolve atomic steps at the surface of Si, Au and CaIrS_4 and image the 7×7 surface reconstruction of Si(111) (Figure 1). They were awarded the Nobel Prize in Physics in 1986. However, a limitation of STM is that the sample needs to be conductive. In 1986, Binnig *et al.* developed the first atomic force microscope (AFM), in which the STM probe (a metallic wire) was replaced by a cantilever made of a diamond glued onto a strip of gold, and the deflection of the cantilever was monitored by tunneling a current between a wire and the gold lever. Rapid improvement in the design of the cantilever and the

methods for monitoring deflection led to the availability of the first commercial AFMs in 1988. Since then, AFMs have evolved to be easier to set-up, to be used on a wider range of materials, and to probe more properties of the material at the nanoscale. As you may imagine the cantilever is no longer a home-made diamond glued to a gold strip—most cantilevers are now fabricated from Si wafers. Similarly, the deflection of the cantilever is no longer monitored by tunneling. We will go on to discuss the design of modern AFMs in the next Chapter.

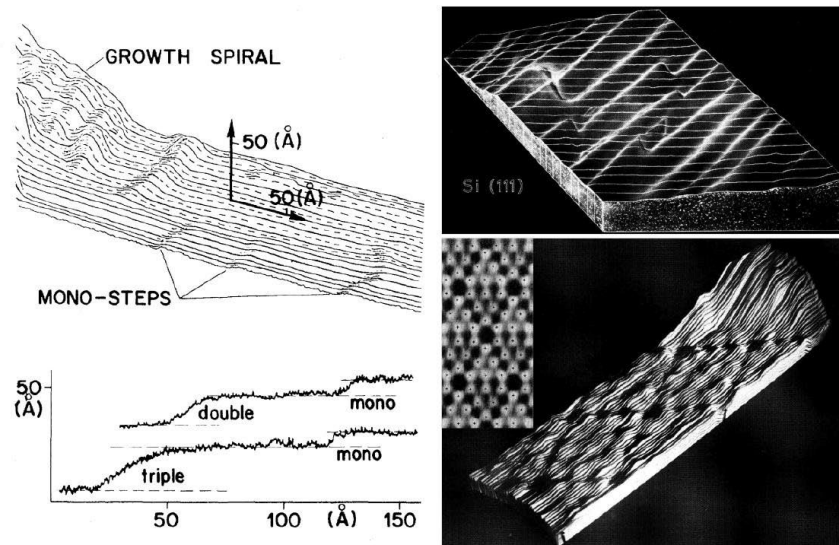


Figure 1: The first STM images: (left) atomic steps on $\text{CaIrSn}_4(110)$ surface (from Binnig et al. *Phys. Rev. Let.* **49**, 57 (1982)); (top-right) atomic steps on $\text{Si}(111)$ surface and (bottom-right) 7×7 surface reconstruction of $\text{Si}(111)$ (from Binnig et al., *Helvetica Phys. Acta* **55**, 726 (1982); *Phys. Rev. Let.* **50**, 120 (1983)).

2 AFM basics

A typical architecture of a modern AFM is shown schematically in Figure 2. Perhaps the most important part of the microscope is a sharp needle (the tip) attached to a flexible cantilever. The horizontal and vertical motion of the cantilever is controlled by a piezoelectric actuator. As the tip is rastered across the surface of the sample, changes in tip-sample interaction can deflect the cantilever. The deflection of the cantilever is monitored by a laser that reflects from the cantilever towards a photodiode. Usually a feedback circuit keeps the tip-sample interaction at a fixed value.

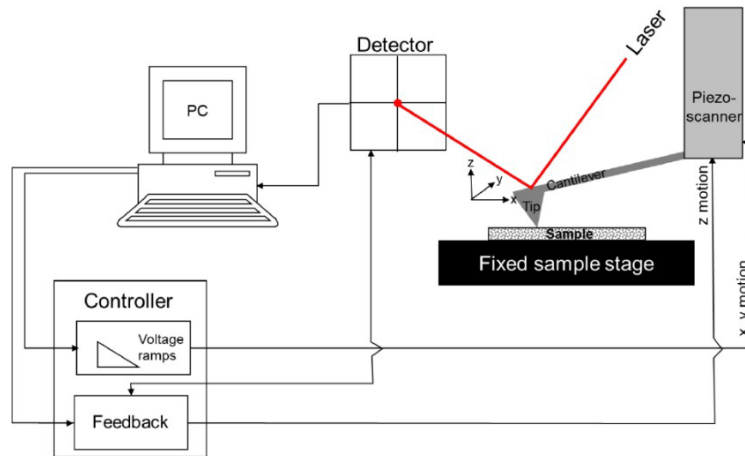


Figure 2: Schematic of an AFM.

But what do we mean by tip-sample interaction, and what are the forces that are applied to the tip? We can think about this in terms of the Lennard-Jones potential, which approximates the interaction between a pair of neutral atoms or molecules – here the atom at the apex of the tip and an atom at the sample surface. This is illustrated schematically in Figure 3, and involves two main forces:

- The Coulombic force is a repulsive force caused by the electrostatic interaction of the electrons in the tip with the electrons in the sample.
- The van derWaals force is an attractive force arising from the interaction between the atoms

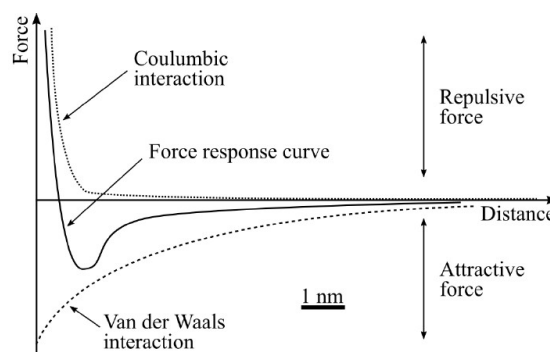


Figure 3: Schematic illustration of the Lennard-Jones potential which models the interaction between two neutral atoms as they approach one another.

from the tip and the atoms from the sample. Its effect is longer range than the Coulombic force.

As well as these fundamental forces, one needs to remember that the AFM is often operated in air. This implies that the sample (and the tip, too) is covered by a thin layer of moisture. This results in adhesion between the tip and the sample.

Using those ideas, we can understand what happens when the tip approaches the sample surface and is then pulled away again. Figure 4 summarises this interaction.

- In position 1, the tip approaches the surface, but the separation is so high that none of the aforementioned forces apply.
- In position 2, the tip gets close enough to the surface to be attracted towards the surface under the dominant influence of the van der Waals force.
- When the force gradient, dF/dz , exceeds the cantilever spring constant, the tip snaps into contact with the sample (the jump-to-contact).
- In position 3, the tip is in contact with the surface, and cantilever deflects according to Hooke's law (i.e. linear dependence).
- Upon withdrawal of the tip, the behaviour of the cantilever initially follows the same trajectory as during the approach.
- In position 4, if a fluid layer is present, a meniscus forms between the tip and the sample surface. Adhesion maintains the tip in contact with the surface.
- Once the adhesion force is overcome, the cantilever breaks contact with the sample (the jump-off-contact)

Different modes of AFM utilize different parts of this force-displacement curve and will be explained below.

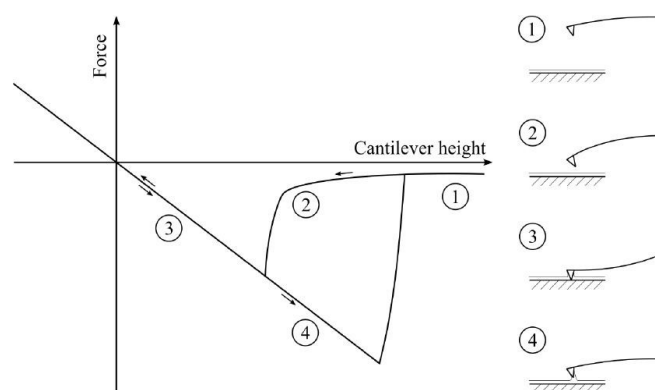


Figure 4: Schematic of force-displacement curve, which describes the forces the cantilever experiences as the tip is engaged and retracted from the sample surface.

2.1 Contact mode

In contact mode, the tip remains in contact with the sample - we are in the linear section of the force-displacement curve. As the tip is scanned across the sample surface, the deflection

of the cantilever is monitored. In the simplest version of contact mode (without feedback), the cantilever is allowed to deflect when it encounters a change in height. The deflection would then be measured for each pixel of the image, and the surface topography would be deduced from it. However, large changes in topography would result in large forces, which could damage the tip.

To overcome this limitation, a feedback circuit in the microscope is used to maintain a fixed deflection of the cantilever (i.e. a fixed force), called the deflection setpoint. In this version, when the system detects that the deflection of the cantilever deviates from the deflection setpoint, the feedback circuit sends a signal to the z-piezo actuator to change the height of the cantilever so that the deflection returns to the setpoint value.

2.2 Tapping mode (intermittent contact mode)

In tapping mode, the cantilever is driven to oscillate at a frequency ω_a , which is at (or slightly below) its resonant frequency ω_r . Typical oscillation amplitude ranges from 20 nm to 100 nm. The tip lightly taps the sample surface at the bottom of each oscillation - on the force-displacement plot the tip typically oscillates between non-contact regime and the linear, contact regime. As the tip-sample

distance varies, the resonant frequency changes according to $\omega_r = \omega_0 \sqrt{1 - \frac{1}{\kappa} \frac{dF}{dz}}$ where ω_0 is natural frequency of oscillation in free space (far from the sample), and κ is the cantilever spring constant. Because the cantilever is driven at a fixed frequency, changes to its resonant frequency result in changes to the amplitude of vibration. In addition, and more importantly, here, the amplitude of the oscillation is damped when the tip is closer to the surface (Figure 5).

Similarly to contact mode, a feedback circuit is used. In this mode the amplitude of the oscillation is used as a setpoint—the amplitude setpoint.

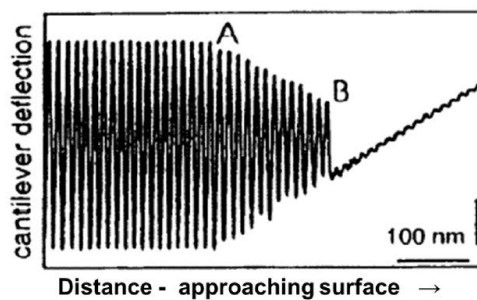


Figure 5: Dependence of the amplitude of oscillation of the tip with distance from the sample. Far away, the tip oscillates at a constant amplitude. From A to B, the amplitude decreases linearly with distance to the sample. Beyond position B, the cantilever energy cannot overcome adhesion and the tip is “stuck” to the sample (From Burnham et al. *Nanotechnology* 8, 67 (1997)).

2.3 Non-contact mode

In non-contact mode, the cantilever is also driven to oscillate at a frequency ω_d , which is at (or slightly below) its resonant frequency ω_r . Unlike in tapping mode, the tip ideally does not contact the surface – it remains in the attractive, van der Waals regime on the force-displacement curve. Here, long-range forces affect the cantilever resonant frequency according to $\omega_r = \omega_0 \sqrt{1 - \frac{1}{\kappa} \frac{dF}{dz}}$. As before, the shift in resonant frequency also causes the oscillation amplitude to vary. Like for tapping mode,

the feedback circuit also operates with an amplitude setpoint. One of the common issues with non-contact mode is that the fluid layer is thicker than the range of the van der Waals force gradient. Therefore, non-contact imaging is most effective when conducted in a vacuum, where it can be used to minimize the forces acting on the tip and the sample.

2.4 PeakForce tapping mode

For the last ten years, peakForce tapping mode has been implemented in most microscopes. Here, the tip is brought into contact with the sample and withdrawn by an oscillation of the main z-piezo. These “oscillations” have an amplitude of typically 100 nm. Note that in tapping and non-contact modes, the oscillation is at resonant frequency ($\sim$ tens of kHz), whereas in PeakForce tapping it is few kHz. Every oscillation follows a whole cycle on the force-displacement curve. In this mode, it is more convenient to see the force-displacement curve as a force-time curve (Figure 6). A change in surface height will affect the time at which the maximum force is applied in the cycle (and also the magnitude of this force). Here, the feedback circuit operates with a peak-force setpoint and aims for the peak-force setpoint to always have the same magnitude and occur at the same time in the cycle.

The advantage of peak-force tapping mode is that one has much greater control of the force applied to the tip (down to pN), which can then stay sharper for longer—we will see later that this is critical to the lateral resolution. Also, the whole force-displacement curve can be analysed, and other information about the sample can be extracted (see later: Quantitative Nanomechanical Mapping).

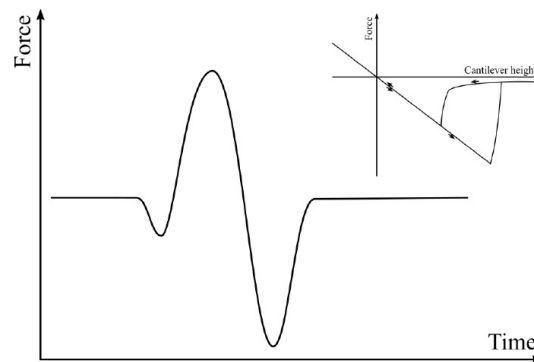


Figure 6: Schematic of force-time curve, which describes the forces the cantilever experiences as the tip is engaged and retracted from the sample surface.

2.5 Error signals

For any of the above modes, the error signal corresponds to the deviation of the tip-sample interaction (deflection/amplitude/peak-force) from the setpoint value. This error signal is recorded during data capture alongside the topography. Visualisation of the error signal can be useful to assess of how well the feedback system maintains the desired setpoint value. An optimal feedback will minimize the error signal—i.e. the system reacts quickly and accurately to a change of surface height. The image formed by the error signal can also complement the height image. For example it can help the user to visualise high spatial frequency changes in height in an image where there is already a broad range or gradual change in surface height. This is because broad range or gradual change in surface height often do not contribute significantly to the error signal but make the topography image difficult to interpret. On the other hand, high spatial frequency change in surface height contributes more to the error, making them more visible in the error signal (Figure 7). (It should be noted that although an error signal image highlights steep gradients on the sample, it is not physically a gradient image. An image of the gradients present in the sample can be achieved by taking the derivative of the height data).

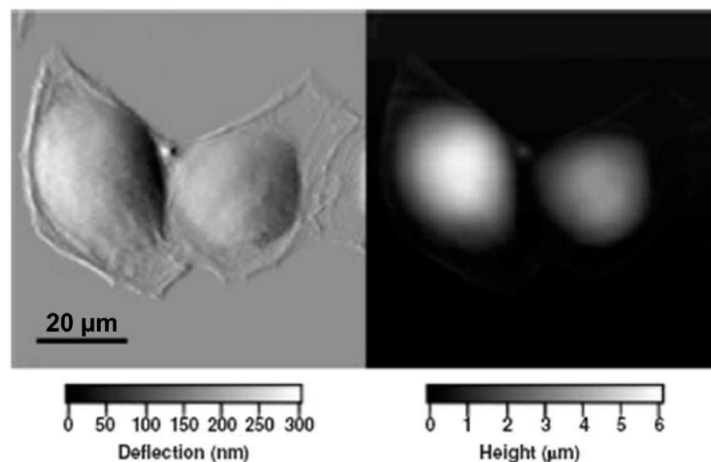


Figure 7: Conjoined cells imaged by AFM in contact mode. The deflection image (left) reveals small sharp features on the cell that do not appear easily on the height image (right). (From Matzke et al. *Nat. Cell Biol.* **3** 607 (2001))

2.6 Summary of the image formation mechanism

For all the imaging modes we have discussed, the same image formation mechanism is relevant. For each pixel in an image:

- The detector senses deviation of cantilever deflection/oscillation/force from the setpoint value.
- The feedback circuit responds to this deviation.
- A bias is thus applied to the z-piezo to adjust the height of cantilever to that the setpoint value is recovered.
- This applied bias is converted into a change in height using calibration data.
- This process is repeated for all pixels in an image as the tip is rastered over the surface.

Although these points are general for different modes, you should now understand that different modes sample different regions of the force-displacement curve. These differences are summarized in (Figure 8). The pros and cons of the different topographic measurement modes are summarized in the table below Figure 8.

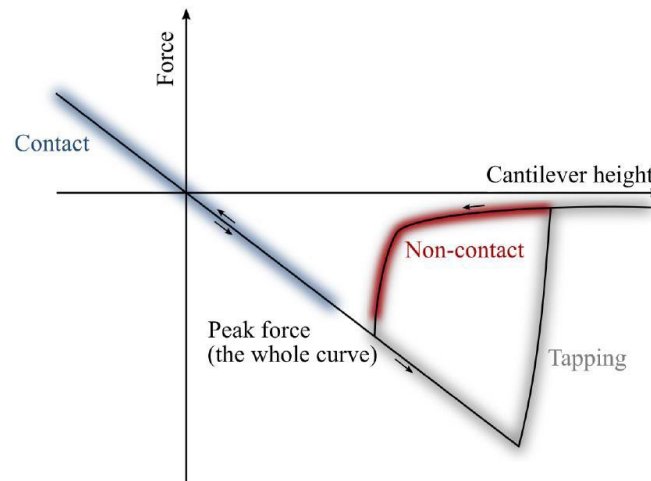


Figure 8: Schematic of force-displacement curve, highlighting the region of operation of each mode.

AFM mode	Advantages	Disadvantages
Contact	- Higher scan speed - Easier mode to setup on rough samples	- Lower resolution - High normal and lateral forces - May damage soft samples
Tapping	- Higher lateral resolution - Reduced normal forces - Nearly no lateral forces - Less damage for the sample	- Lower scan speeds
Non-contact	- Minimal normal and lateral forces - Atomic resolution in UHV	- Limited resolution in air (fluid layer) - Slowest scan speed
Peak force	- High lateral resolution - Good control over normal forces - High stability (noise sensitivity) - Force curve can be analysed	- Lower scan speeds

3 Components of an AFM

In this Chapter, we will learn in a bit more detail about the key components of the microscope, as these will help you understand how to set it up and how artefacts in the image can arise (Chapter 4).

3.1 Feedback

The previous Chapter emphasized that the feedback circuit is a central element of the microscope. The setup of the feedback circuit determines the ability of the system to track the sample surface accurately—and the topographic images are then obtained from the height at which the z-piezo places the tip during the scan.

The feedback circuit is a PID system (PID stands for proportional-integral-derivative). Without going into too much detail, the underlying idea is that the system corrects deviation from the actual setpoint value (the proportional term), deviation from the setpoint value over time (the integral term), and rate of deviation change (the derivative term). In practice, it means that you will have to find the correct values for the feedback gains, which are numbers which determine the sensitivity of the feedback circuit to each of the proportional, integral and derivative terms. A higher gain implies the feedback circuit is more sensitive to deviations in that term.

The feedback gains are manually adjusted while scanning by monitoring the height and error profiles (Figure 9).

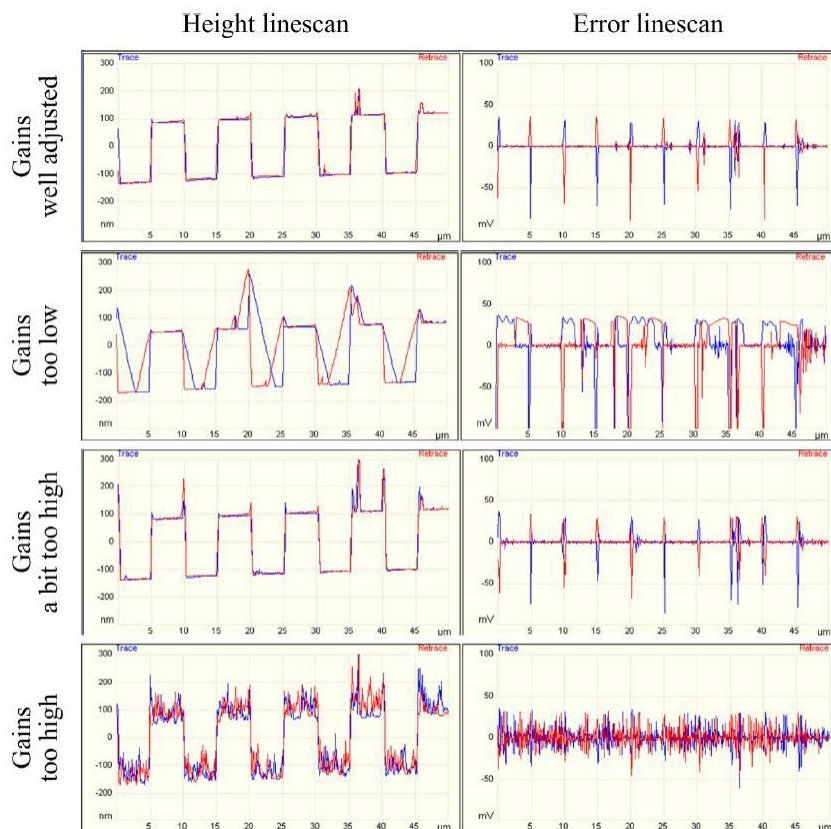


Figure 9: Height and error linescans of a square profile with (from top to bottom) gains well adjusted, gains set too low, gains set a bit too high, and gains set too high.

- When the feedback is well-adjusted, the forward and reverse (trace and retrace) lines in the height data match perfectly, and the amplitude of the error signal is minimised.
- When the gains are set too low, the system is too slow at tracking the surface (large errors are required for the feedback to kick in). The trace and retrace in the height linescan do not match - in particular, one may see the tip “ski-jumping” (losing contact with the surface) as the surface height goes down.
- When the gains are set too high, the system becomes unstable and oscillates. Noise in the system will cause random deviations from the setpoint that will be amplified by the feedback. It results in a noisy scan. Moreover, one may see the system over-correcting changes in height, for example, overshoots when the surface height goes up.

3.2 Piezoelectric scanner

The scanner is made of a piezoelectric material (often PZT, $\text{Pb}_x\text{Zr}_{1-x}\text{TiO}_3$), which expands or contracts under an applied voltage. The piezo scanner is usually shaped as a hollow tube, with several sections allowing movement in the x, y and z direction (Figure 10 (left)). Note again here that the scanner does not only provide the x-y raster motion which allows scanning of the surface, but also responds to voltages arising from the action of the feedback circuit to move the tip up and down, maintaining the tip-sample interaction at the setpoint value. One has to bear in mind that the movement of the piezoelectric scanner is not linear with applied voltage – hysteresis and creep occur.

- Hysteresis corresponds to the different path taken by the scanner upon expanding or retracting. To obtain a linear movement of the scanner, a non-linear voltage waveform must be applied (Figure 10 (top right)). This must be calibrated regularly (to account for the aging of the piezoelectric material) for a range of scan sizes.
- Creep occurs when a large voltage is applied to the scanner. Upon application of a step change in voltage, the scanner motion is not completed instantaneously. Instead, the scanner

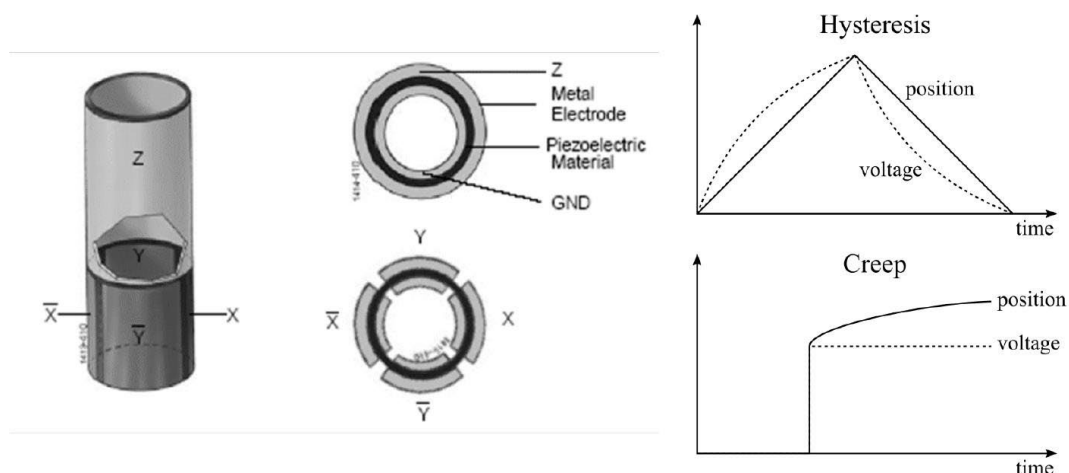


Figure 10: (left) Schematic of a hollow tube piezo actuator. Representation of non-linearity phenomena: (top-right) hysteresis, and (bottom-right) creep.

may continue to move slowly in the same direction for some time after the new voltage level has been applied. (Figure 10 (bottom right)).

3.3 Tip and cantilever

AFM imaging is a convolution of the sample surface with the probe geometry. The geometry of the tip (end-radius, sidewall angles) strongly influences the quality of the image generated - this will be discussed further in Chapter 4.

The design of the cantilever itself needs to be adjusted for the mode of operation—the spring constant and resonant frequency are particularly important. For example:

- In contact mode, the cantilever is designed to have a low spring constant, to obtain large deflections from small forces, and high resonant frequency to avoid vibrational instabilities. Such cantilevers therefore tend to be short and thin.
- In tapping mode, on the other hand, a high spring constant is required to reduce noise and instabilities.

3.4 Photodetector

The cantilever deflection is monitored by the position of a laser beam reflected from the cantilever back side on a four-sector photodetector. The deflection of the cantilever, as used in contact mode, is given by the differential signal between the top and bottom quadrants of the detector. In addition, in lateral force microscopy (discussed later), the difference in signal recorded between the left and right quadrants is also used.

3.5 Mechanical isolation

Given the mechanical nature of the AFM imaging process, the microscope is extremely sensitive to external vibrations. That is why it is crucial to isolate the microscope: for example, the departmental AFM is mounted on an air table and is covered by an acoustic hood. Even with these precautions, loud speech or a door banging can affect the microscope performance. Moreover, the 50 Hz electrical supply and other electrical signals can also give rise to observable noise in the image.

4 Resolution and Artefacts

4.1 Resolution

- Vertical resolution: The absolute limit of the vertical resolution is determined by the resolution of the z-piezo scanner movement which is $< 1 \text{ \AA}$. However, digitisation of the data (pixelation) and noise actually limit the resolution in practice—the best achievable resolution in the departmental AFM is ca. $0.3 \text{ \AA}$.
- Lateral resolution: The lateral resolution of the microscope is primarily determined by the tip geometry—especially the radius of curvature of the end of the tip. To give a rough idea, a lateral resolution of 5-15 nm is frequently achievable, but it could get worse with tip wear, or better (ca. 2 nm) with sharp tips used in PeakForce tapping mode.

Although AFM can provide atomic resolution (laterally), this is in practice extremely challenging. The influence of the finite tip radius and the presence of the moisture layer in ambient air make it almost impossible, and the vast majority of atomic force microscopy imaging is achieved in ultra-high vacuum. It should be noted that even in ultra-high vacuum, the fact that van der Waals force operate over a relatively long range (in comparison to the tunneling current in STM), makes it difficult to exclude the impact of the tip sidewalls from the data, degrading resolution. STM is more usually applied for atomic resolution imaging of conductive materials, but non-contact AFM techniques in ultra-high vacuum are increasingly employed for imaging non-conductive materials or individual molecules. It is also worth noting that when imaging in a fluid environment, no water droplet forms around the tip, because the whole cantilever is already surrounded by fluid. Hence, imaging in fluid can also yield surprisingly excellent lateral resolution.

4.2 Tip artefacts

We mentioned earlier that the image obtained by AFM is a convolution of the surface with the tip. Indeed, the geometry of the tip plays an important role in the resulting topography - it determines the lateral resolution of the microscope. First, one needs to understand that the tip is not infinitely sharp - the very apex of the tip during imaging can typically be modelled as a sphere of typical radius 5-15 nm (Figure 11 (top)). In peak force tapping mode, where there is a better control of the force applied to the tip, sharper tip radii are often maintained with a nominal end radius of 2 nm. The tip also has sidewalls which angles can affect the image too.

Tip end radius: The finite curvature of the tip implies that every raised feature at the surface of the sample will be broadened, and that the tip cannot necessarily penetrate fully into surface pits and trenches (Figure 11 (bottom)). The impact of the finite tip radius is particularly important when

imaging small, or high aspect ratio features. It's also important to be aware that a worn tip or a blunt tip will have a larger radius, resulting in a loss of resolution, and often requiring a new tip to be installed.

Tip sidewalls: When imaging features with a high aspect ratio, it is common that the surface is probed by the sidewalls of the tips instead of the apex of the tip (as in the schematic at the bottom of Figure 11). It should be noted that the sidewalls of the tip can have different inclination angles (front, back and side angles)—which should be specified by the manufacturer. Hence rotating the scan direction will affect the observed sidewall angle steep-sided features. If steep-sided features need to be imaged accurately, very high aspect ratio probes are available, but these are often expensive and fragile.

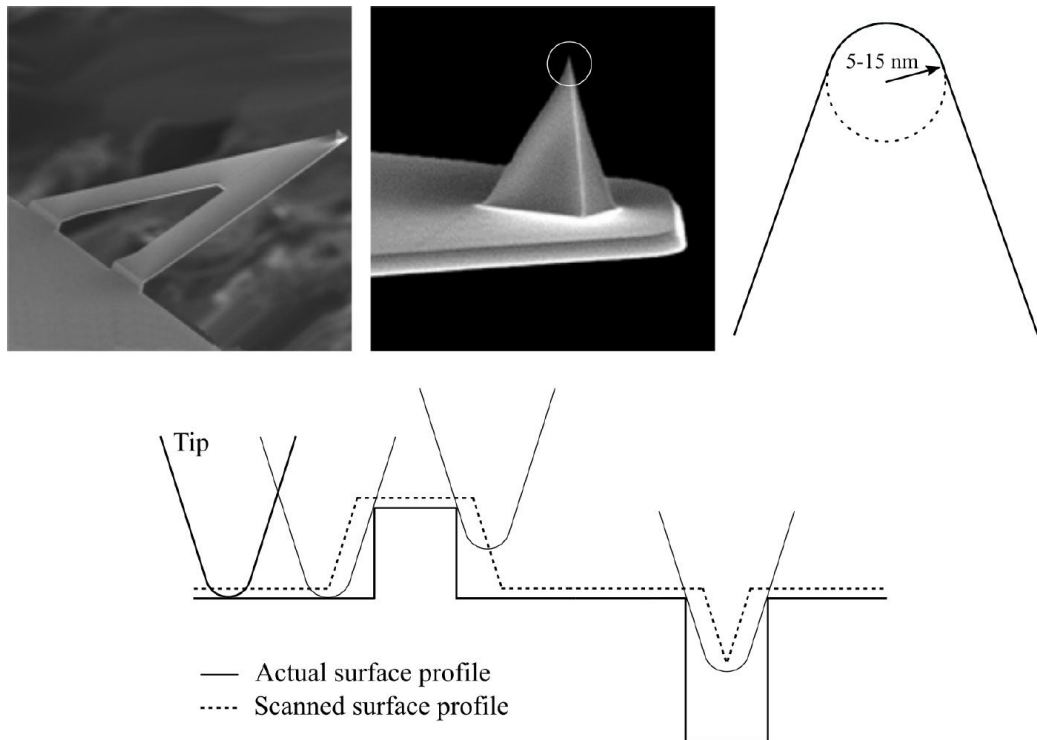


Figure 11: (top) Scanning electron microscopy image of a tip (from www.brukerafmprobe.com). (bottom) Impact of tip geometry on topography.

Contaminated or broken tip: An AFM tip is not everlasting—it is a consumable. AFM tips sometimes break, for example from mishandling or scanning tall features too fast. Another scenario, which leads to the same image artefact, is that the tip get contaminated, for example by picking up a piece of debris on the sample's surface. In both cases the sample is scanned by a probe which has a shape which can be quite different from the ideal tip shown in Figure 11. A symptom of this problem in the AFM image is the appearance of “unusual” features which are always oriented the same way (Figure 12). If you suspect a broken tip artefact, take another image by rotating the sample (not the scan direction!). If the features still have the same direction, then it is a tip artefact and the tip needs changing.

Double-tip: Double-tip artefacts are a variant of the contaminated or broken tip artefact. As the name indicates, the damaged probe in this case has two (or more) tips which can interact with the sample.

The features at the surface of the sample are not necessarily distorted, but are duplicated. There is always a fixed orientation relationship between the multiple images of the same feature. However this artefact is sometimes sample dependent - e.g. in some cases a smooth sample may not exhibit the artefact, while a rough sample would (Figure 13). Hence, this artefact can sometimes be a bit tricky to spot. If double-tip artefact is suspected, apply the same procedure as for contaminated or broken tip.

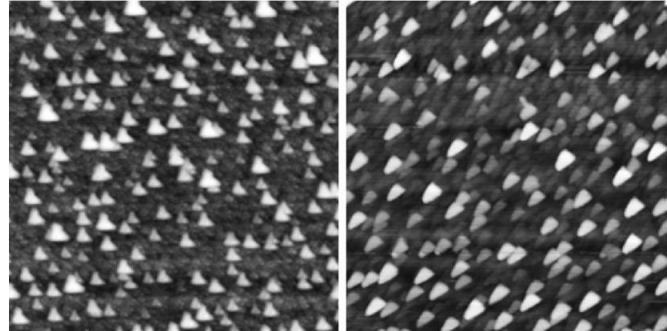


Figure 12: Examples of broken tip artefacts (from www.nanophys.kth.se).

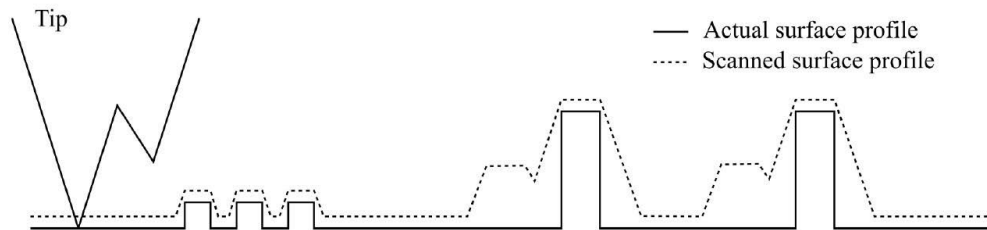
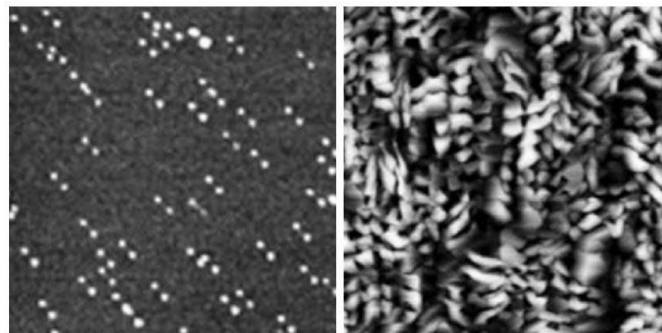


Figure 13: (top) Examples of double-tip artefact (from www.nanophys.kth.se). (bottom) Impact of double-tip on topography.

Tip changes: Tip changes can occur when the tip has picked up some contamination. This may not lead to a significant degradation in resolution. However, when the tip picks up a debris from the sample surface, the end of the debris can be the effective probe interacting with the sample surface, and the presence of the debris effectively makes the tip longer. The system understands this as a sudden change in the height of the surface. This artefact can be easily removed using imaging processing called “flattening” (discussed later).

4.3 Scanner artefacts

The piezoelectric scanner also induces artefacts.

Bow: Because the piezoelectric scanners are fixed at one end, with the tip attached to the other, moving end, the free end does not move in a level plane. The mechanical properties of the piezo as well as the kinematics of motion may result in 2nd or 3rd order deviation from the ideal plane. This artefact is known as “bow” (Figure 14 (left)), and increases with scan size. This artefact can be removed by flattening the image (see later).

Creep: We mentioned earlier that creep occurs when a voltage is applied to the piezo, which continues moving even after the voltage has been applied. The image will therefore be distorted by this residual motion (Figure 14 (right)). This is very noticeable when a large offset (10s of μm) has been applied. The main way to eliminate creep is to wait for the motion of the piezo to stabilize. Many modern scanners can also be operated in a closed-loop mode, which uses an additional feedback to correct these non-linearities. Unfortunately, this tends to increase the noise level in the scan.

Closed-loop is not generally advisable for small scan sizes where very high resolution is required, but can be very helpful for larger scan sizes.

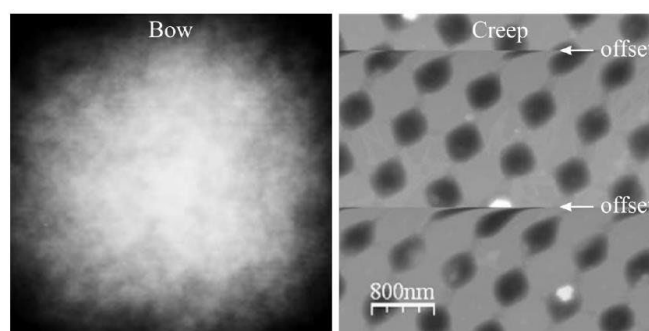


Figure 14: (left) Example of bow (from www.nanophys.kth.se). (right) Example of creep (from www.doitpoms.ac.uk).

Sample drift: Although this is not a scanner-related artefact, sample drift is listed here because its appearance in the AFM image can be rather close to creep. Sample drift can occur when the sample is not well fixed down, or when the sample is not in thermal equilibrium with its surroundings. Whilst creep gives an undesired component of tip motion relative to the sample, drift gives an undesired component of sample relative to the tip. Both result in distorted (stretched or squashed) features, but drift recovers much more slowly than creep. If the sample is drifting, it is possible to minimize the impact of this problem by rotating the scan direction so that the most important profile of the sample features is scanned along the fast scan direction.

4.4 Periodic noise

Periodic noise can come from electrical or mechanical sources. It appears as periodic fringes on the AFM image, whose spacing does not depend on the scan size (but depends on the scan rate). This artefact can be removed by image processing, using Fourier transform techniques. This and other image processing techniques will be discussed in the next chapter.

A table summarizing all the common artefacts is included as an appendix at the end of the handout.

5 Image processing and data analysis

AFM data in its raw form may need to be processed to provide an accurate representation of the surface – and a pretty picture for presentation. However, AFM is not only about acquiring pretty pictures; it can be used for quantitative surface morphology measurements. This section gives a non-exhaustive list of the most common data processing and analyses that could be performed on an AFM scan.

5.1 Analysis software

Analysis of AFM images can be performed using the manufacturer's software which is provided with each microscope. However, different microscopes use different softwares, and hence a preferred option is often to use free software, available online, which can be generally applied to data from microscopes from several different manufacturers:

- Gwyddion: Available at <http://gwyddion.net/>. (Quite powerful, but not too user friendly).
- WSxM: Available at <http://www.wsxm.es/>. (More user friendly, however functionality a bit more limited - but more than enough for what we discuss here).
- Nanoscope Analysis (by Bruker): Installed in the departmental AFM. Not openly available but a copy can be obtained upon request.

5.2 Flattening

Flattening is an image processing technique applied almost ubiquitously to AFM data. As mentioned in the previous chapter, flattening is used to remove scanner bow and tip change artefacts, both of which occur very frequently. However, if your image already looks great and flattening is not needed, you should avoid doing it because it results in information loss.

Briefly, flattening takes every line of the AFM image (in the fast-scan direction). It fits the height profile along that line with a polynomial function (which could be a simple offset or a 1st or 2nd order polynomial – i.e. a straight line, or a parabola). This fitted function is then subtracted from the original profile. For a 0th order polynomial, the “fitting” process is simply the calculation of an average. That average is subtracted from every point on the scan line, hence bringing the average height of each line to zero. Since this is done for every fast-scan direction line, height information in the slow-scan direction is removed.

For this simple example, a key element of the information loss is in the slow scan direction. If different lines taken in the fast scan direction should (in an accurate data set) have different average heights, then the flattening process removes this difference. This is illustrated in Figure 15, which also shows that the appropriate orientation of sample features relative to the fast scan direction can help reduce the damaging impact of flattening.

When flattening non-uniform surfaces—especially when these contain isolated deep or tall features—flattening will generate horizontal streaks (Figure 16). These are due to the feature shifting the average

height of the line in which it lies up or down. These streaks are an artefact of the image processing. It is possible to use the analysis software to prevent these streaks by selecting features with anomalous heights to be disregarded by the flattening algorithm.

Most microscopes use real-time flattening to help with visualisation, but save the raw data to avoid the loss of information inherent in flattening. Flattening is often necessary to remove artefacts from the image, but should be applied intelligently, not habitually.

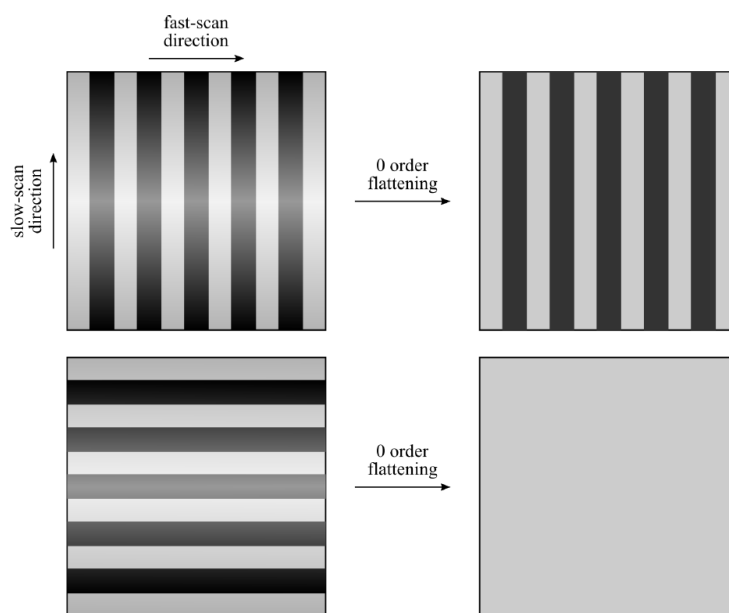


Figure 15: *Flattening does not lose much information in the fast-scan direction (top), but does in the slow-scan direction (bottom).*

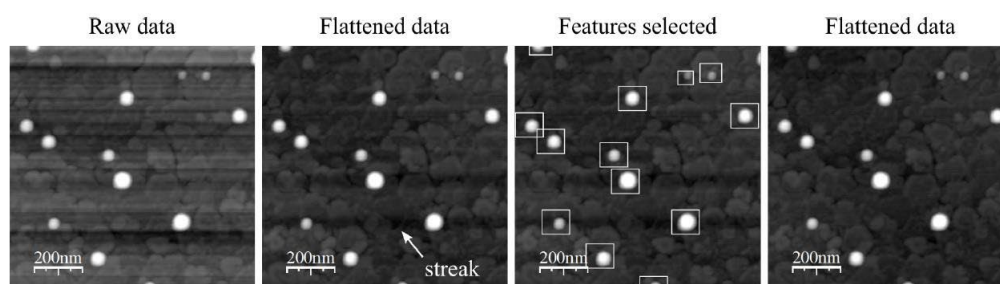


Figure 16: *(from left to right) Raw unflattened AFM scan; AFM image when flattening is applied over the entire scan, resulting in horizontal streaks; Selection of features that cause the streaks; AFM image when flattening is applied outside the selected features.*

5.3 Streak removal

Horizontal thin (1 or 2 lines) streaks can be present in the AFM image as a result of a noisy environment during scanning—a classic cause is door banging. These can be removed automatically (in Gwyddion software) or manually (in WSxM software). The algorithm simply takes out the faulty line and replaces it with an average of the adjacent lines. A few streaks on a scan are not a big deal. However, if you have too many of them, it will be visually obvious that the image has been corrected. In that case, consider re-taking the image.

5.4 Removing periodic noise

Fast-Fourier transform techniques can be used to remove periodic noise from images. Given that this noise is by definition periodic, it will have a clear signature in reciprocal space (i.e. in frequency domain). One can then remove the corresponding region of the Fourier transform of the image, and reconstruct the image by inverse Fourier transform without the contribution of noise. In general, Fast-Fourier transforms are a means to highlight periodical features in the image and analyse the direction and average size of those features, and a number of much more sophisticated analysis routines for AFM images based on this approach may be discovered in the literature.

5.5 Measuring heights, depths and widths

Having processed your data to remove artefacts and achieved a data set which best represents the original surface, you are in a good position to acquire quantitative morphological data. A classic thing the AFM allows us to do is measure dimensions, laterally and vertically. However, measuring height and width of surface features accurately is not always as straightforward as it sounds. Here we describe some rigorous methods for data analysis.

Measuring step height: Many samples exhibit a terraced morphology, where it may be interesting to measure the height of the steps (recall the first STM images (Figure 1) with atomic steps!). However, suppose your data have been flattened using a 1st order polynomial or above. In that case, you may not see a series of steps as a staircase, but instead see the surface as being on average horizontal, since the overall slope of the surface is removed by the flattening process (Figure 17 (left)). This prevents accurate measurement of the step height. We need horizontal terraces with vertical risers in between. A local plane fit can be applied by selecting several regions of the surface that are supposed

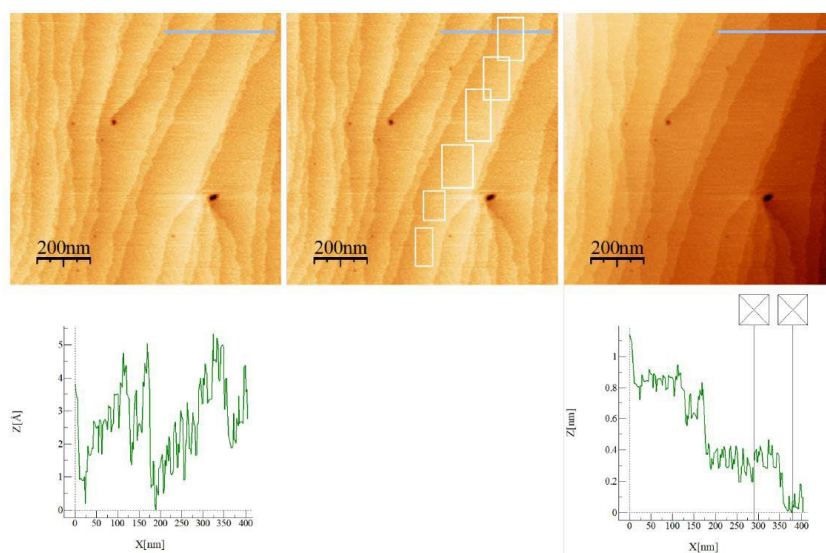


Figure 17: (left) Flattened image of a terrace structure, with height profile. (middle) Area from a same terrace are selected for local plane fitting. (right) Resulting image and height profile.

to lie on a same horizontal plane (selected areas in Figure 17 (middle)). The resulting topography is now a proper staircase (Figure 17 (right)) and accurate step height measurements can be undertaken.

Measuring hills and troughs: Characterisation of hills and troughs can also be a tricky thing to do.

First of all, always bear in mind that the tip affects the width of a feature and the depth of pits. Then,

the material surrounding the feature may not have the same height (due to flattening, for example) on both sides of the feature—a local plane fit, if relevant in this sample, may help. Once an adequate reference height has been found, measuring the height of a hill should not be a problem. Measuring its width, on the other hand, leaves more room for arbitrary interpretation. A common approach to measuring width is to quote the full-width at half maximum. One must be a bit more careful characterising pits as this is even more affected by the tip status. It is not recommended, for example, to give depth values unless you are certain that the tip can reach the bottom of the pit, instead, quoting measurements of width may be more appropriate—using the “full-width at half minimum” or better, measuring the width at a fixed distance below the surrounding surface (the full-width at half minimum requires knowledge of the depth, which is particularly dependent on the tip status).

Irrespective of the chosen method, the most critical factor here is that you are consistent! If possible, try to use the same tip for all measurements to alleviate uncertainties from tip status.

5.6 Global measures of surface roughness

Very often, you may be interested in characterising the AFM image in its entirety, rather than focusing on individual features. The analysis software can readily give a range of more or less useful metrics, allowing you to compare samples. Here we will review the most common ones.

Z-range: This is the difference in height between the highest and lowest points in the image. This is not a good global measure of surface roughness since only 2 pixel of the image are involved! However, it is a useful measure of the z-scale of an image, and you may want to state it in a figure legend when displaying the data (e.g. “image height = XX nm”). A better approach may be to include a Z-scale bar, however.

Image mean: The image mean describes the mean height over the image. It is rarely useful: if your data haven’t been flattened at all, the average height tells you more about the artefacts and the offset from the arbitrary experimental zero level than about the sample; if your data have been flattened by setting the mean of every line to zero, then the image mean will necessarily be zero!

Roughness: The roughness is probably the most useful (and most commonly used) global metric to describe the sample topography. Put simply, it describes spread in height. Two common roughness

measures are used: the average roughness, $R_a = \frac{1}{N} \sum_{i=1}^N |Z_i - \bar{Z}|$, where N is the number of pixels, Z_i is the height of the pixel i , and $\bar{Z}$ is the mean height; and the root mean square (rms) roughness,

$R_q = \sqrt{\frac{1}{N} \sum_{i=1}^N (Z_i - \bar{Z})^2}$, R_q is essentially the standard deviation of height. The main difference

between R_a and R_q is that R_q is more sensitive to very high or very low features. When trying to compare samples based on their roughness, one has the following points to bear in mind:

- The roughness does not tell you everything about your sample. Different topographies can give the same roughness.
- A roughness value is only valid for a specific image size. You should always quote the image size alongside the roughness. Consequently, comparison between samples is only meaningful if you use the same image size.
- Flattening affects the roughness – higher-order flattening decreases the roughness. Similarly, comparison between samples is only meaningful if they undergo the same image treatment.
- The roughness depends on the tip status. The variation of roughness with tip size is not a straightforward function, however, and depends on the nature of the surface topography. More details in Chen *et al.* Meas. Sci. Technol. 15, 2005 (2004).

6 Electrical measurements by AFM

So far, we have only addressed using AFM to measure the topography of a sample. However the AFM can also measure a wide range of materials properties, provided that it can detect a suitable interaction between the tip and the sample. Here, we review the most common techniques for electrical characterisation of a sample by AFM.

6.1 Lift Mode

Many AFM electrical property measurement techniques based around tapping mode utilise the so called “lift mode”. This mode allows the imaging of relatively weak but long-range magnetic or electrostatic interactions while minimising the influence of topography. In lift mode, every line of the sample is scanned twice. On the first pass, the “main” scan line, topographical data is recorded in tapping mode in both the trace and retrace directions. Then, the system raises the tip to the lift scan height. Next, the system performs a second trace and retrace scan (the “lift” scan line) while maintaining a constant tip-sample separation. The tip-sample separation is maintained by using the topographic data from the previous scan line to control the tip height. During this second scan, the electrical or magnetic data is recorded. Lift mode is used for magnetic force microscopy (MFM), electric force microscopy (EFM) and Kelvin probe force microscopy (KPFM).

6.2 Magnetic force microscopy

In MFM, the tip is coated with a ferromagnetic thin film to be sensitive to the magnetic properties of the sample. The magnetic field between the tip and the sample depends on the tip-sample separation and the magnetic properties of the sample. The MFM signal is recorded during the lift scan. During this scan line, the tip-sample distance remains constant, and the cantilever oscillates at (or near) its resonant frequency. As mentioned in Chapter 2, the resonant frequency is a function of the vertical force gradient: $\omega_r = \omega_0 \sqrt{1 - \frac{1}{k} \frac{dF}{dZ}}$. Changes in the magnetic properties of the sample will thus affect the resonant frequency of the cantilever. Sensitivity to these changes in resonant frequency can be achieved by:

- Amplitude detection: Measuring the shift in amplitude at fixed drive frequency.
- Phase detection: Measuring the shift in phase at fixed drive frequency.
- Frequency modulation: A feedback loop modulates the drive frequency to keep the cantilever’s phase lag at 90° relative to the drive frequency, corresponding to resonance.

The interpretation of MFM images is complex because the magnetic force depends on the magnetic structure of the tip as well as the sample. When imaging soft magnetic samples with an MFM tip that has a high magnetic moment, displacement of the sample’s domain walls may occur during scanning.

Conversely, if the magnetisation of the sample gives rise to strong stray fields, this may lead to rotation of the magnetisation field of the tip, altering the image.

The achievable resolution depends on the tip's domain magnetisation, shape, length, cone angle and distance from the surface: in general, MFM resolution is roughly equal to the lift height, and magnetic features smaller than the lift height may not be resolved. The tip also experiences stronger fields close to the surface, improving signal-to-noise ratios.

6.3 Electrical force microscopy

In electrical force microscopy (EFM), the tip is coated with a conductive film, and a DC bias may be applied. EFM works on the same principle as MFM, except that in the EFM case, it is electrostatic, rather than magnetic, forces that are probed. The same detection methods may be applied. EFM can map the electrostatic fields of an electronic circuit as the device is turned on and off. This technique is known as “voltage probing” and is a valuable tool for testing live microprocessor chips at the sub-micron scale.

6.4 Kelvin probe force microscopy

In KPFM, as opposed to EFM and MFM, the cantilever is not vibrated during the lift scan. Instead, an AC voltage is applied to the tip $V = V_{ac} \sin(\omega t)$ which creates an oscillating electric force at the frequency ω on the cantilever. The oscillating force has the following amplitude: $F = \frac{dC}{dz} \Delta V_{dc} V_{ac}$ where $\frac{dC}{dz}$ is the vertical derivative of the tip-sample capacitance and $\Delta V_{dc} = V_{tip} - V_{sample}$. When the tip and sample are at the same DC voltage ($\Delta V_{dc} = 0$), the cantilever experiences no oscillating force. Hence, by using a feedback circuit to adjust the DC voltage on the tip (V_{tip}), until the oscillation amplitude equals zero, the effective local surface potential on the sample (V_{sample}) may be determined. The surface potential image is then a map of V_{sample} over the surface (Figure 18).

Unlike in EFM and MFM, the cantilever oscillation amplitude should be close to zero. (The feedback circuit aims to zero the oscillation). Hence, the tip may be scanned closer to the surface (at a smaller lift height) without the tip impacting the sample. Lift heights as low as -5 nm are sometimes used (i.e. 5 nm lower than the mean height during the main scan line). If the cantilever does impact the surface, streaks appear in the surface potential data. However, one must be careful as sample topography can affect KPFM measurements. The amplitude of oscillation of the cantilever is proportional to the

oscillating force, hence $A = k \frac{dC}{dz} \Delta V_{dc} V_{ac}$. The feedback circuit aims to keep the amplitude of

oscillations to zero. However, noise in the system prevents this, which means that there is a residual vibration of amplitude A_{min} . This implies that the tip bias must be given by: $V_{tip} = \frac{A_{min}}{V_{ac} k \frac{dC}{dz}} + V_{sample}$,

the term $\frac{A_{min}}{k \frac{dC}{dz}}$ represents an error in setting the tip potential equal to the sample potential which

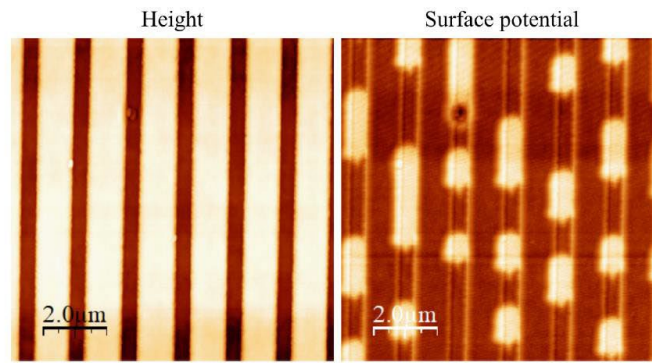


Figure 18: Topography (left) and surface potential (right) images of a CD-RW. The surface potential image shows the position of the written bits on the tracks. (Courtesy of Rob Cork.)

depends on the tip-sample interaction area because it depends on $\frac{dC}{dz}$. (The capacitance, C , of the tip-sample combination, depends on their separation and the area of the capacitor which they form (more details on Efimov et al., J. Vac. Sci. Technol. B 18 1051 (2000)). In conclusion one should be wary if variations in surface potential match variations in topography.

6.5 Scanning capacitance microscopy

In scanning capacitance microscopy (SCM), a metal-coated tip is in contact with a semiconducting sample with an oxide layer on its surface. The set-up then forms a metal-insulator-semiconductor (MIS or MOS) capacitor. An AC bias is applied between the tip and the sample, and the capacitance value is monitored using a high frequency resonant circuit (Figure 19 (left)). The microscope is operated in contact mode, maintaining a constant force between tip and sample, allowing simultaneous topographic and capacitance images to be generated. This enables the direct correlation of local topography with electrical properties.

Let's take the example of an n-type semiconductor. With a positive voltage applied to the tip, electrons in the semiconductor are attracted to the surface and accumulate there. The capacitance is then the capacitance across the oxide layer. For a negative tip voltage, the electrons are repelled from the tip, depleting the material of carriers. This effectively increases the spacing between the capacitor plates and lowers the capacitance. The lower the carrier concentration the greater the decrease of capacitance with voltage. Therefore, the slope of the C-V curve (i.e. $\frac{dC}{dV}$) larger for lower carrier concentrations

(Figure 19 (right)). For p-type material, the C-V curve polarity inverts. Therefore, the amplitude of the SCM signal is characteristic of the dopant concentration (but care is required in data interpretation: high amplitude means low dopant concentration), while the phase of the data (essentially a measure of which half of the voltage cycle leads to a capacitance change) can be used to distinguish n- and p-type semiconductors. However, a calibration structure is needed to quantify the dopant

concentrations. SCM also allows for spectroscopic measurements, where site-specific —

$\frac{dC}{dV}$ versus V curves can be taken. These curves can provide extra insight into the material's properties at specific locations.

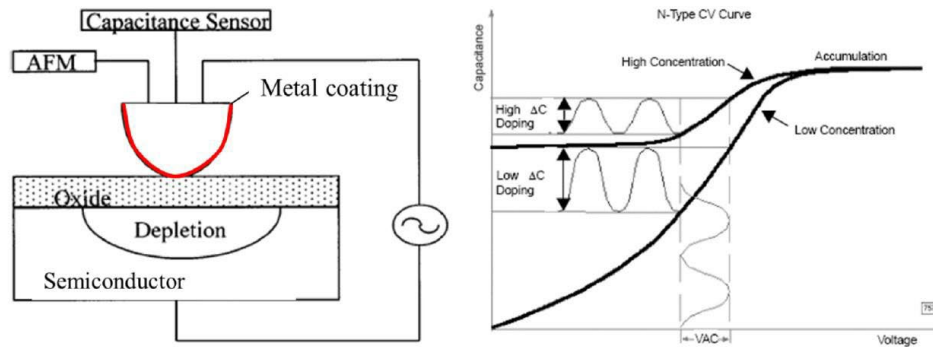


Figure 19: (left) Schematic of SCM set-up. (right) C-V curve for an n-type semiconductor with high and low carrier concentration.

6.6 Techniques involving tip-sample current flow

6.6.1 Scanning spreading resistance microscopy

Scanning spreading resistance microscopy (SSRM) measures the resistivity of samples spanning a wide range of conductivity, from insulating through semiconducting to metallic. A DC bias is applied between a conductive tip and the sample. While scanning in contact mode, a logarithmic current amplifier with a typical range of 10 pA to 0.1 mA senses the current, I , passing through the sample. Simultaneous topographic and current images may be generated, enabling the direct correlation of local topography with electrical properties.

In semiconductors with a native oxide layer, strong forces are required to penetrate through the oxide layer and establish a stable electrical contact. That is why SSRM often employs Si tips coated with doped-diamond, or even (very expensive) all-diamond pyramids. SSRM also allows for spectroscopic measurements, where site-specific I - V curves can be taken. As with SCM, calibration structures are often used to quantify dopant densities. On the other hand, unlike SCM, SSRM does not differentiate between n- and p-type materials unless they have different resistivities.

6.6.2 Conductive AFM and tunnelling AFM

C-AFM and TUNA operate in a similar way to SSRM, but the output is conductivity rather than resistivity. These techniques use a linear rather than a logarithmic current amplifier. The difference between the two linear techniques is their sensitivity: C-AFM can sense currents from about 1 pA to 1 μA, while TUNA can typically sense currents in the range 80 fA to 120 pA. TUNA got its name because it was originally designed to estimate the thickness of (insulating) oxide layers via the measurement of current tunnelling from a tip to an underlying conductor or semiconductor. (Note that the tunnelling current here flows through an insulating layer, which is different to STM – see later – where the tunnelling current flows across a vacuum gap).

Like SSRM, C-AFM and TUNA allow for spectroscopic measurements, where site-specific I - V curves can be taken. It should be noted that SSRM, C-AFM and TUNA are techniques that are very sensitive to the tip-sample contact area. This depends on the tip size but also on the sample

morphology. Hence, be careful not to over-interpret results when the conductivity matches the topography.

Note: In some recent microscopes, which allow for peak-force tapping imaging, TUNA can now be conducted in peak-force tapping mode, with current only flowing when the tip is briefly in contact with the sample at the bottom of each oscillation.

6.7 Piezoresponse force microscopy

In piezoresponse force microscopy (PFM), the tip is scanned over the surface in contact mode and an AC bias is applied. The *average* deflection of the tip is used for topography measurements. The phase and amplitude of the AC component give the piezoresponse of the sample and are collected by a lock-in amplifier, allowing imaging of ferroelectric domains.

7 Mechanical measurements in AFM

7.1 Friction force microscopy

In friction force microscopy (or lateral force microscopy), the scanner moves perpendicular to the long axis of the cantilever. Friction between the tip and the sample causes the cantilever to twist, and the four-quadrant photodetector can distinguish the left-right motion of the laser from the up-down motion induced by the topography. In this mode, topography and friction can be recorded simultaneously.

However, the topography also affects the twisting of the cantilever, because the lateral force is high at the leading edge of asperities and low at the trailing edge (Figure 20). Moreover, the magnitude of material-induced effects should be independent of the scan direction whereas topography-induced effects are different between the trace and retrace scan directions. Since the sign of the friction force changes when the scanning direction is reversed, the addition of the friction force data of the forward and reverse scan should eliminate material-induced effects while topography-induced effects remain (Figure 20).

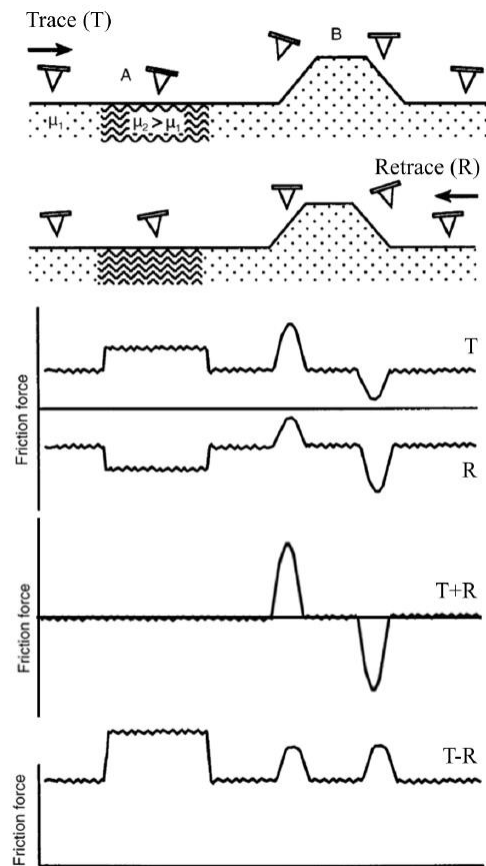


Figure 20: Working principle of friction force microscopy (from Bhushan, *Wear* 259, 1507 (2005)).

7.2 Phase imaging

As we have previously noted, the resonant frequency of the cantilever in tapping mode varies as $\omega_r = \omega_0 \sqrt{1 - \frac{1}{Q^2}}$. However, in discussing tapping mode imaging, we noted that the damping of the

oscillations with proximity to the surface were the dominant mechanism which allowed the cantilever oscillation amplitude to be used as a measure of surface height. However, the phase, ϕ , of the cantilever's vibration relative to the phase of the driving oscillation also varies as $\tan(\phi) =$

$\frac{\omega_d \omega_r}{Q(\omega_d^2 - \omega_r^2)}$, where ω_d is the frequency of the driving oscillation, and Q is the quality factor of the

resonant system. Hence, the phase of the cantilever could give us access to extra information about the forces acting on the tip. In practice, the phase signal is affected by variations in composition, adhesion, friction, viscoelasticity, and perhaps other properties.

Applications of phase imaging include identification of contaminants, mapping of different components in composite materials, and differentiating regions of high and low surface adhesion or hardness.

For the phase signal to be completely unaffected by the sample topography, the tip-sample distance would have to be kept perfectly constant. As we know, the feedback circuits are not perfect and require finite time to respond to height changes. Hence, phase shifts will arise due to the changes in the force gradient as the tip-sample separation changes. This is particularly noticeable for topographic contributions with high spatial frequencies, and phase imaging can be used to highlight these components on samples with both high and low spatial frequencies.

7.3 “Quantitative” nanomechanical mapping

We mentioned in Chapter 2 that in peak force tapping, the tip goes through an entire force-displacement curve cycle to obtain the surface height. However, much more information about the sample is contained in the force-displacement curve (Figure 21); the most common metrics to extract from the curve are:

- *Dissipation*: Dissipation describes how much energy is spent in cycling the tip in and out of contact with the sample.
- *Adhesion*: Adhesion describes how much force is required to pull the tip out of the sample.
- *Deformation*: Deformation is a measure of how much the tip penetrates into the sample.

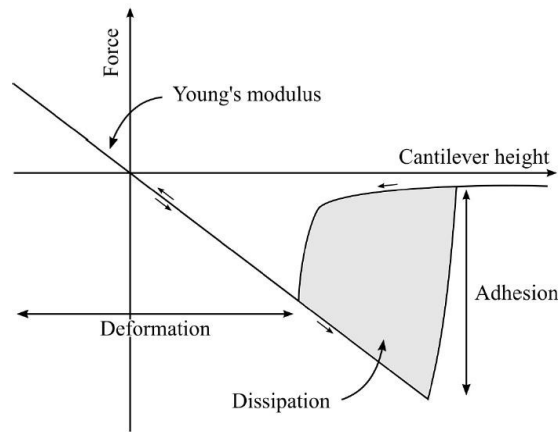


Figure 21: Summary of the information that could be extracted from the force-displacement curve.

- *Young's modulus:* Young's modulus can be calculated if the elastic properties of the tip, its shape, and Poisson's ratio are known. Various models of the deformation occurring at the tip may be used in this calculation, and the user must be aware of the limitations of such models. Often, not all the necessary parameters are accurately known, even if an appropriate model is being applied.

Quantitative nano-mechanical imaging is a variant of peak force tapping where maps of the various mechanical data outlined above are extracted concurrently with the topographic data.

Note that, despite what the name may suggest, this technique is, in practice, often *qualitative*, but that can be sufficient to compare the stiffness of various features within an image. Proper quantification is sometimes possible but can be very time-consuming.

7.4 Nanoindentation

The tip of the AFM can be used as a small indenter. For nanoindentation in the AFM, the tip is lowered into contact with the sample, indented into the surface, and lifted off the sample surface. The deflection is recorded as a function of the vertical displacement of the piezo scanner to know the applied force, and the surface may be imaged in tapping mode immediately before and after indentation.

Tips for nanoindentation are fabricated by precision grinding of solid diamond. These tips are sufficiently sharp for imaging in tapping mode after indentation. Each cantilever is individually qualified and supplied with an image of an indentation—this is important to obtain interpretable results.

7.5 Nanofabrication

Having introduced the possibility of nanoindentation, we note that plastically deforming the sample (indenting it) is a simple form of nanofabrication. A number of other more sophisticated AFM-based nanoindentation techniques have been developed. The list below is certainly non-exhaustive but gives a flavour of applications of AFM for nanofabrication:

- *Scratching*: An extension of the application of the AFM tip as a nanoindenter is the deliberate fabrication of structures by scratching the surface. As a nanofabrication method this is fairly limited due to the influence of tip wear and the amount of debris produced on the surface.
- *Resist exposure*: The field emission current from a biased AFM tip may be used to expose e-beam resist. The process resolution depends on the resist thickness, and the cross-sectional shape of the developed resist pattern is dictated by the amount of exposure (Ishibashi et al., Appl. Phys. Lett. 72, 1581 (1998)).
- *Selective oxidation*: AFM tip-induced local oxidation is an electrochemical process based on negatively biasing the tip concerning the substrate under ambient conditions. Since the oxidation process requires an electrical current, both the tip and substrate must have some conductivity - commonly used substrates include metals or semiconductors that form stable oxides. The negatively biased tip induces a high electric field, which ionizes the water molecules from the ambient humidity between the tip and the substrate. This produces OH⁻ ions, which act as an oxidant. The electric field enhances the vertical drift of these ion species away from the tip towards the surface, where they react with underlying atoms to form a localized oxide beneath the tip.
- *Dip-pen nanolithography*: An AFM tip is coated with a thin film of ink molecules that react with the substrate surface to write nanoscale patterns. In a high-humidity atmosphere, a nanoscale water droplet condenses between the AFM tip and the substrate. The drop of water acts as a bridge over which the ink molecules migrate from the tip to the substrate surface, where they are deposited (Piner et al., Science 283, 661(1999)).
- *Structural transformation*: Some materials can undergo structural transformation under compression and/or charge injection. AFM is well/suited to altering the properties of the material at the nanoscale, where a given force is applied to the material, and a bias is applied between the tip and the sample (Barboza et al., Adv. Mater. 23, 3014 (2011); Barboza et al., ACS Nano 12, 5866 (2018)).

8 AFM in biology

Because AFM can be operated in fluid (buffer solution), this technique is suitable for the study of biological samples. Let's first start with practical considerations. A first challenge for imaging in fluid is that the biomaterial must be immobilised onto a substrate—often mica, glass or quartz—the method for that depends on the material which is being imaged:

- *Biomolecules*: As simple adsorption is often insufficient, the surface can be treated, or the pH or ionic composition of the buffer fluid could be altered to enhance the binding of the biomolecule to the substrate. Alternatively, a monolayer that would form a covalent bond with the molecule could be attached to the substrate.
- *Cells*: Many animal cells tend to spread and adhere to solid surfaces naturally. Coating the substrate with an adhesion protein such as collagen may enhance immobilisation. Alternatively, cells can be fixed in place using a suction pipette or immobilised mechanically in an agar gel or porous membrane.

The fluid ambient environment could also vary—from static to dynamic conditions:

- *Static conditions*: Imaging can be performed in a liquid droplet deposited on the substrate, although this is limited to hydrophobic substrates. Furthermore, liquid evaporation may be an issue. Alternatively, imaging in a liquid bath alleviates these disadvantages and allows for better temperature stability; for microscopes not explicitly designed for in-fluid operation this can be more difficult to set-up. These days, specialist microscopes intended for biological applications are available where imaging set-ups in a liquid bath or other fluid-containing cell are relatively facile.
- *Dynamic conditions*: Dynamic conditions, where the sample is immersed in a liquid bath with an inlet and outlet, allow imaging of the cell under changing buffer conditions. However, drag forces arising from the moving liquid may affect the tip motion, especially given that low spring constant cantilevers are used for biological imaging (to avoid damaging the sample).

One of the key applications of AFM in biology (other than imaging) is the measurement of force-displacement curves in which a folded biomolecule is anchored at one end to a substrate and attached at the other end to the tip. As the tip moves away from the sample, the molecule is placed under tension and unfolds, and the force (or energy) required to achieve different unfolding events is measured (Figure 22).

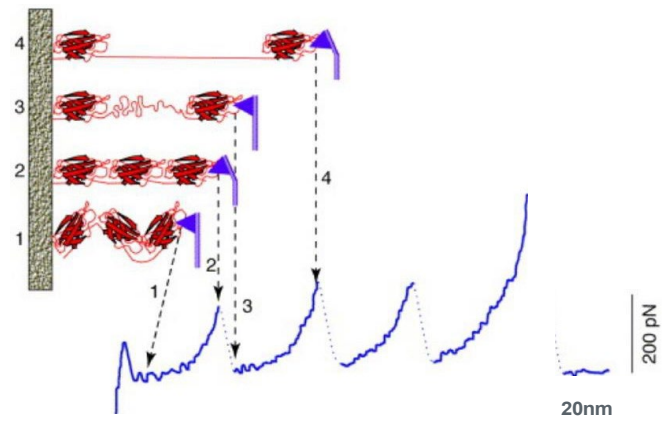


Figure 22: Saw-tooth pattern of peaks that is observed when force is applied to extend the protein corresponds to sequential unravelling of individual domains of a modular protein (from Fisher et al., *Trend Biochem. Sci.* **24**, 379 (1999)).

9 Scanning tunnelling microscopy

9.1 STM basics

In scanning tunnelling microscopy (STM), a metallic tip is positioned a few angstroms away from a conductive sample. A bias is applied to the tip which allows some electrons to tunnel through the potential barrier between the tip and the sample.

In a basic scenario, illustrated in Figure 23, the tunnelling current is approximated by $I \propto e^{-2Kd}$ where $K^2 = \frac{2m}{(\hbar/2\pi)^2} \frac{\phi_{tip} + \phi_{sample}}{2}$. Here ϕ_{tip} and ϕ_{sample} are the work function of the tip and sample, respectively. We can see that the tunnelling current is exponentially dependent on the tip-sample separation, which makes it extremely sensitive to changes in height. In STM, atomic resolution of metals and semiconductors can be achieved almost routinely because of this high sensitivity – tunnelling occurs from a single atom at the tip apex to the nearest surface atom, and other tip atoms are hardly involved since they are further from the surface.

However, the basic mathematical model above assumes that atoms are hard spheres. Tunnelling involves electron transfer between states at the Fermi level of the tip and states at the Fermi level of the sample. We need to take into account the local density of states (LDOS). The Tersoff-Hamman model was the first attempt to include the influence of the tip shape and surface electronic density of states in the tunnelling formalism (Tersoff *et al.*, Phys. Rev. Lett. 50, 1998 (1983)).

This model assumes that the tip is spherical and can therefore, be described by spherically symmetric s-wave functions only. It also assumes low voltages and low temperatures. This model suggests that the tunnelling conductance is proportional to the LDOS of the sample surface at the Fermi level evaluated at the centre position of the tip. Hence, a constant current image of the surface maps out a contour along which the surface LDOS at the Fermi level is always constant.

The Tersoff-Hamman model predicts a resolution of 6-9 Å and works well on some samples, like Au(110). However, it fails to describe STM measurements on semiconductor samples, where the low voltage assumption does not apply. Moreover, resolutions of 2-4 Å can be achieved experimentally, which is better than predicted by theory. This is because the s-wave model does not incorporate the geometry or electronic structure of the real tip. In fact, localised p_z or d_{z^2} tip states are actually responsible for the atomic resolution of the STM (Chen *et al.*, Phys. Rev. Lett. 65, 448 (1990)).

Nonetheless, the basic insight of the Tersoff-Hamman model – that the STM image represents a contour of constant LDOS – underpins the majority of basic image interpretation used for STM data. More sophisticated image interpretation tends to involve atomistic modelling, and although this is significantly beyond the scope of this course, we will touch on this approach briefly in section 9.3.

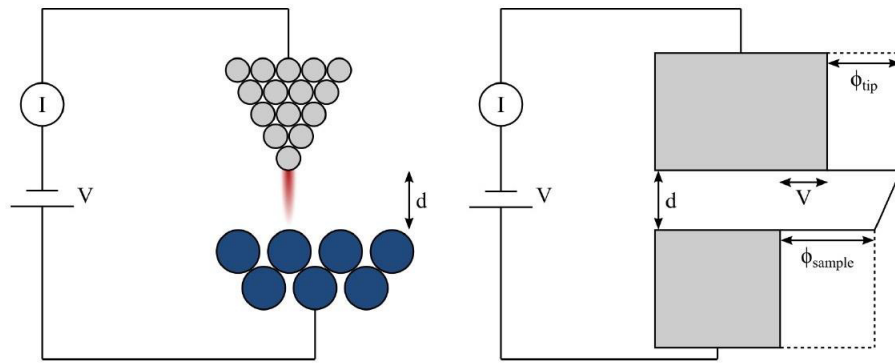


Figure 23: (left) Schematic of STM set-up, and (right) corresponding band diagram.

9.2 STM instrumentation

Whilst tunneling is possible in air, most STM experiments are performed in ultra-high vacuum (UHV) (at pressures $< 10^{-9}$ Torr) as this allows surfaces to be kept clean. Excellent vibration isolation is also required. since the aim is to achieve an accuracy in height of ca. $0.01 \text{ \AA}$. Achieving both UHV and low vibration noise simultaneously requires using pumps with no moving parts, known as ion getter pumps—the basic idea is to ionise the residual gasses in the vacuum and then adsorb the ions to the wall of the pump. Piezo-elements and control electronics similar to those used in AFM are employed. In contrast with AFM, the tip shape has little impact in STM because the sample is essentially probed by a single atom located at the apex of the tip. This is due to the exponential dependence of the tunnelling current with tip-sample separation. Typical tip preparation methods include mechanical cutting (poor aspect ratio), electrochemical etching (better aspect ratio), and in-situ heating or field evaporation.

Unlike in TUNA, the aim of STM is not usually to understand the structure of a surface oxide (by tunnelling through the oxide), but to tunnel through a vacuum gap onto a clean (oxide-free) surface to understand the atomic structure of that clean surface. However, many samples (especially semiconductors) have a native oxide, which must therefore be removed before imaging.

The STM can be operated in a number of modes, the most common of which are noted below:

- *Constant current mode:* A constant current equates to a constant tip-surface “distance”. A current setpoint is supplied by the user, and the feedback circuit acts to maintain that setpoint. Hence, the tip-surface distance is kept roughly constant, and the voltages applied to the z-piezo can be calibrated to give changes in “height” (here “distance” and “height” should be understood in the context of LDOS).
- *Constant height:* The feedback loop is turned off, and the tip scans at a constant height over the surface. Variations in the tunnelling current are digitized and used to make up the image.

High scanning rates may be achieved in this way (e.g. for the observation of dynamical processes), but it is much more likely that you will crash (and damage) the tip in this mode.

- *Scanning tunneling spectroscopy*: With the tip stationary, I vs V or $\frac{dI}{dV}$ vs V curves may be recorded. Peaks in the $\frac{dI}{dV}$ vs V curves are interpreted as corresponding to energies with a high LDOS (e.g. surface states or resonances). If we want to take a spectroscopic image, then with the feedback circuit engaged, the tip-sample separation changes when the voltage is changed (to maintain the setpoint current). Hence, topographic and electrical information may be convolved. CITS (current imaging tunnelling spectroscopy) deals with this problem by alternating during the scan between the feedback circuit on (for topography measurements) and the feedback circuit off (for I - V measurements).
- *Atomic manipulation (vertical)*: STM can be used to move individual atoms vertically. An atom is transferred from the surface to the tip by bringing the tip close to the atom and applying a voltage pulse. The tip (with the atom) is then lifted from the surface and moved laterally. Next, the tip is brought close to the surface and the atom is deposited (using a voltage pulse). The voltage's polarity determines the direction in which the atom moves (on to or off the tip).
- *Atomic manipulation (lateral)*: STM can be used to move individual atoms laterally. The adsorbate is kept adsorbed on the surface; the tip is brought down towards it, forming a weak bond. The tip is then moved along the surface, dragging the adsorbate. The tunnelling current can be used to monitor tip height during the manipulation process.

9.3 Image interpretation

Interpretation of STM atomic resolution images is complicated, as indicated already by our discussions in section 9.1. Here are a few key ideas which you should be aware of:

- *Surface reconstruction*: The creation of a surface means that the coordination of the atoms and the electronic structure in the surface region is changed relative to the bulk. As a result, the atoms move relative to their positions in the bulk crystal. For many metals, this movement is only a slight relaxation (expansion or contraction), which will minimise the surface free-energy. For some metals (e.g. Au, Pt and Ir) and most semiconductors, the surface can lower its energy significantly by a significant rearrangement of atoms known as a reconstruction. For compound semiconductors, the equilibrium (lowest energy) reconstruction will depend on the stoichiometry of the surface. Sub-monolayer quantities of surface adsorbates also give rise to different reconstructions.
- *Voltage-dependent imaging*: When the sample is negatively biased, electrons tunnel from the occupied states of the sample to the unoccupied states of the tip. If the sample is positively biased, electrons tunnel from the occupied states of the tip to the unoccupied states of the sample.

Depending on the sample—and on the feature that is observed—this could be used to obtain some chemical contrast. For example, in GaAs, the As sites are electron-rich, whereas the Ga sites are electron-deficient. This means that the Ga sites have a higher density of unoccupied states, and the As sites have a higher density of occupied states. Hence, imaging at negative sample bias (occupied states imaging) reveals the As sub-lattice, whilst imaging at positive sample bias reveals the Ga sublattice (Figure 24).

- Atomistic modelling: Generally, a combination of atomistic modelling and STM imaging is found to be a powerful route to identify the correct surface structure of the sample. The general approach is: generate a plausible model of the surface; solve its electronic structure using, e.g. Hartree-Fock approximation, density functional theory, tight-binding methods or quantum Monte Carlo; compute the resulting STM image (possibly at several biases); compare with experimental data (for more details: Briggs et al., Surf. Sci. Rep. 33, 1 (1999)).

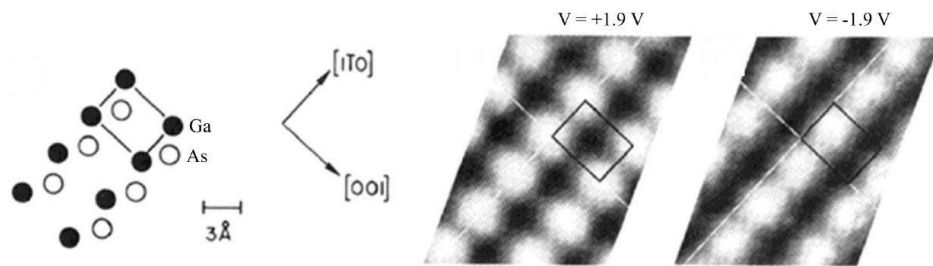


Figure 24: GaAs(110) surface observed by STM under positive and negative bias (from Feenstra et al., *Phys. Rev. Lett.* **58**, 1192 (1987)).

10 Scanning near field optical microscopy

10.1 SNOM basics

Scanning near field optical microscopy (SNOM) exploits the properties of an evanescent wave that forms in the near-field of a sub-wavelength sized aperture. In the optical near-field (distance from aperture $\ll \lambda$) the concept of optical diffraction loses its relevance. Therefore, by placing the sample in the optical near-field region, i.e. very close to the aperture, the light will interact with the sample with a resolution determined by the diameter of the aperture, *not* by the wavelength of the light being used. An image is built up by raster-scanning the aperture across the sample and recording the optical response of the specimen.

Typical probes for SNOM consist of a metal-coated optical fibre with a tapered end and a small aperture in the metal coating at the apex.

AFM and SNOM can be combined to allow measurement of the topography and optical properties of the sample. Moreover, the feedback system of the AFM is useful to maintain a constant tip-sample distance (typically < 20 nm). Most SNOMs use a shear-force feedback mechanism: the probe is vibrated parallel to the surface, and changes in its vibration amplitude with distance from the surface are used to monitor its position.

The microscope can operate in different modes depending on the light illumination and collection pathways.

- *Transmission mode*: The sample is illuminated through the probe, and the light passing through the sample is collected and detected in the optical far field by a standard microscope objective lens.
- *Reflection mode*: The sample is illuminated through the probe, and the light reflected from the sample surface is collected in the optical far field by a standard microscope objective lens.
- *Collection mode*: The sample is illuminated with a macroscopic light source from the top or bottom, and the probe is used to collect the light from the sample surface. I.e. here it is collection that is in the near field, not illumination.
- *Illumination/Collection mode*: The probe is used for both the illumination of the sample and for the collection of the reflected signal.
- *Apertureless SNOM*: This system uses a sharp probe with no aperture (e.g. a standard AFM probe), which is illuminated by a far-field source. The tip locally perturbs the fields at the sample surface, and the far-field response to this perturbation is detected.

10.2 SNOM artefacts

SNOM probes are fairly blunt (compared to AFM probes); hence they may not penetrate crevasses. Relatively small changes in the separation between tip and sample can lead to a large change in the intensity of the light at the sample surface. This may result in variations in light output intensity caused by the probe not being able to follow the topography of the surface and not by changes in the type of material illuminated.

The position of the “optical probe” is effectively defined as the centre of the aperture. However, the “topographic probe” could be a protrusion in the metal coating defining the aperture, which could be some distance from the aperture centre. Hence, an offset may exist between the optical and topographic images.

11 Appendix: Table of artefacts

Name	Cause	Symptoms	Solution
Scanner bow	Scanner does not move in a level plane	False parabolic contrast in x or y direction	Post-processing of image (see lecture 3)
Scanner creep	Large voltage applied to scanner; delayed response.	Features elongated or compressed; recovers with time.	Wait! (Or use a closed loop scanner)
Drift	Changes in scanner temperature; slight changes in sample position	Similar to scanner creep, but generally recovers more slowly.	Check sample is well stuck down. Wait for temperature to stabilise.
Tracking failure	Feedback circuit cannot adjust scanner height quickly enough.	Equiaxed features develop “comet tails”. Forward and reverse scan do not match.	Increase gains, or adjust setpoint to increase interaction, or decrease scan rate.
Feedback instability	Oscillation of feedback circuit	Quasi-periodic noise, especially visible in amplitude or deflection image, and on steeply sloped regions.	Decrease gains
Ringing	Cantilever oscillation increases for both a slight increase and a slight decrease in height	Rings around features; for small features potentially contrast inversion	Adjust the drive frequency (usually reduce it). May need to reduce amplitude setpoint to maintain contact.
Z-pixilation	Digital limitation to sampling resolution in z-direction	Equal sized steps seen in very small features	Decrease Z-limit (i.e. increase sampling resolution by decreasing sampling range).
Damaged tip	Careless use of tips; picking up dirt or debris.	Repeating features across image.	Sometimes possible to knock off dirt by increasing scan size. Usually: change tip.
Double tip	As above	Features occur in pairs (or threes, or fours...)	As above
Skips and streaking	Loose debris on surface; moved by tip.	Long or short white streaks	Clean sample with dry air or nitrogen.
Optical interference	Stray light from laser reflected from shiny surfaces	Broad, quasi-periodic stripes in height image.	Refocus laser on cantilever so less light spills onto sample
Tip changes	Pieces of material adhering to or falling off the tip	Horizontal steps in image.	Clean sample. Post-processing with “flatten” function.
Periodic noise	Mechanical vibrations or electrical interference	Periodic stripes; often particular visible in amplitude or deflection image.	Switch off noise source if possible. Post-processing using FFT.